

$$Av = 34$$

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
First Semester (2024-2025), Mid-Semester Test
Course: Surveying (CE F213)

Date: 4th Oct. 2024

Max. Marks: 75

Duration: 2:00 PM-3:30 PM

- Q.1. The distance between two elevated points with same level is measured with a chain. The distance obtained is 18.5m. This measurement is carried out at a mean temperature of 42°C with a pull of 190N. The cross-sectional area of the chain is $1.256 \times 10^{-5} \text{ m}^2$ and the unit weight of the chain is 1.2N/m. The modulus of elasticity of the chain material is $2.1 \times 10^{11} \text{ N/m}^2$. The coefficient of thermal expansion is 11.0×10^{-6} per °C. The chain used for this measurement was standardized at 20°C with a pull of 100N when supported throughout its length. Find the temperature correction, pull correction and sag correction and their nature. Find the correct distance between the elevated points. Use approximate sag correction formula. [15]

Find normal pull also.

- Q.2. In a levelling operation, the following observations are taken with the auto level. All the reading are in m. While taking the observations, the instrument is changed after 2nd, 7th and 10th readings. Take the RL of the 1st station as 220m.

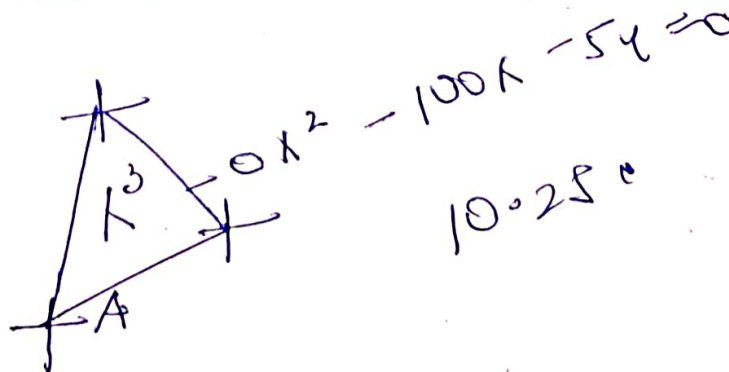
2.215, 1.925, 2.350, 1.550, -1.775, -1.995, -1.685, -1.330, 1.795, 1.985, 1.545 and 1.515

Find the R.L.s of all the stations using rise and fall method. Perform all checks. [15]

- Q.3. The fore bearing and back bearing of the lines in a traverse conducted with the prismatic compass is mentioned in the table. All the bearings are whole circle bearings. The local attraction and observation error are suspected. Find the correct included angles and using those angles find the correct bearings of the lines CD and DE. [20]

Line	FB	BB
AB	100° 45'	281° 48'
BC	135° 18'	316° 10'
CD	45° 12'	224° 14'
DE	300° 36'	122° 39'
EA	260° 53'	79° 18'

- Q.4. In order to find the width of a river, two points A and B were taken on one bank of the river so that the line AB is approximately parallel to the river. A well-defined point C on the other side of the bank was marked. The fore bearing of the line AC was 35° 40' and the fore bearing of the line BC was 355° 24'. Find the width of the river if AB = 195 m. [15]

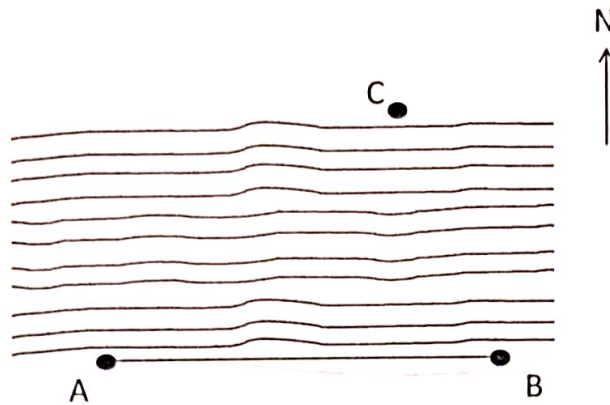


26'

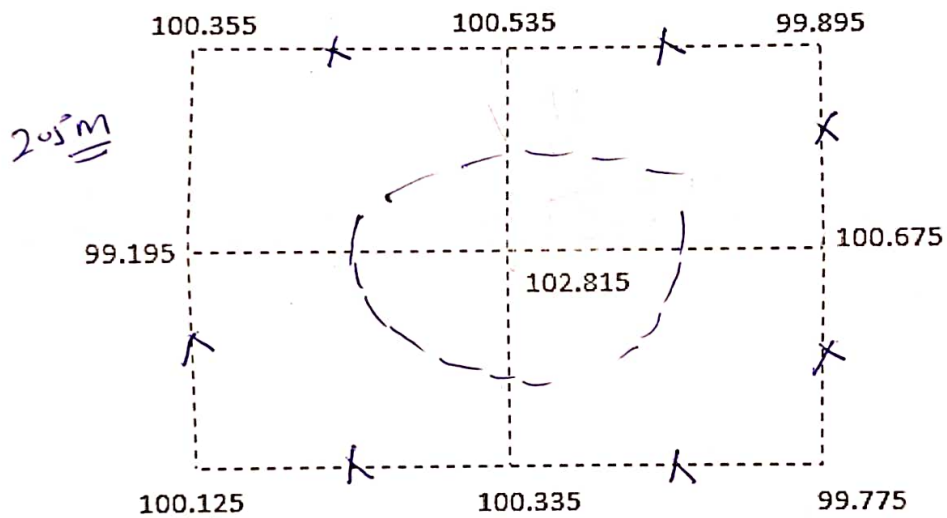
16144'

315° 44'

316° 36'



✓ Q.5. In the following grid (5m×5m) the RL of the points are given in m. A contour of RL 101m is to be drawn. Find the distance of this contour line from the center point. [10]



Q10-1 (i) Temperature correction:

$$C_t = \alpha (T_m - T_0) L$$

$$= 11.0 \times 10^{-6} (42 - 20) \times 18.5 = 0.4477 \times 10^{-2} \text{ m} \\ (+ve) \text{ (3)}$$

(ii) Pull correction:

$$C_p = \frac{(P - P_0) L}{AE} = \frac{(190 - 100) \times 18.5}{1.256 \times 10^{-5} \times 2.1 \times 10^8} = 6.312 \times 10^{-4} \text{ m} \\ (+ve) \text{ (3)}$$

(iii) Sag correction:

$$C_{\text{sag}} = \frac{w^2 L^3}{24 P^2} = \frac{(1.2)^2 \times (18.5)^3}{24 \times 190^2} = 0.0105 \text{ m} (-ve) \\ \text{(3)}$$

Correct distance between the two points is:

$$18.5 + 0.4477 \times 10^{-2} + 6.312 \times 10^{-4} - 0.0105 = 18.4901 \text{ m} \text{ (1)}$$

Normal pull (pull required to make pull correction and sag correction equal)

$$C_p = C_{\text{sag}}$$

$$\Rightarrow \frac{(P - P_0) L}{AE} = \frac{w^2 L^3}{24 P^2}$$

$$\Rightarrow (P - 100) \times 24 P^2 = w^2 L^2 A E$$

$$\Rightarrow 24 P^3 - 2400 P^2 = (1.2)^2 \times (18.5)^2 \times 1.256 \times 10^{-5} \times 2.1 \times 10^8$$

$$\Rightarrow 24 P^3 - 2400 P^2 - 12.999 \times 10^8 = 0$$

$$\Rightarrow P^3 - 100 P^2 - 54.163 \times 10^6 = 0$$

$$P = 414.797 \text{ N} \text{ (5)}$$

No-2 Rise and Fall method.

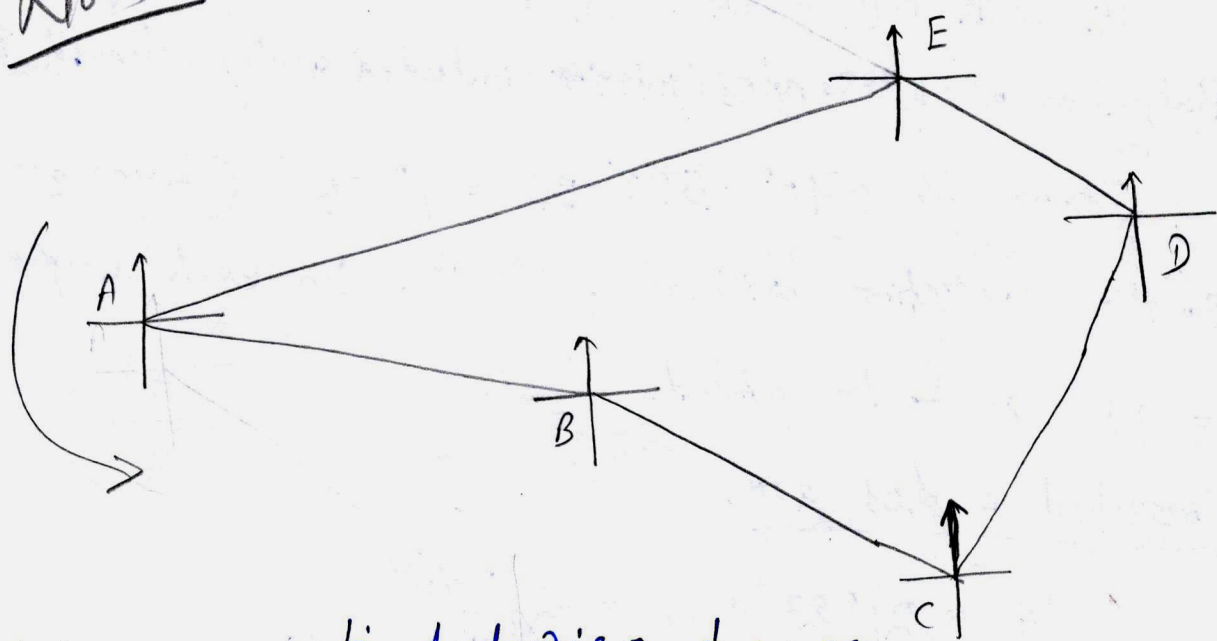
Stn	B.S. (m)	I.S. (m)	F.S. (m)	Rise (m)	Fall (m)	R.L. (m)	
A	2.215					220	B.M.
B	2.350		1.925	0.29		220.29	C.P.
C		1.550		0.8		221.09	
D		-1.775		3.325		224.915	
E		-1.995		0.22		224.635	
F	-1.330		-1.685		0.31	224.325	C.P.
G		1.795			3.125	221.200	
H	1.545	1.985	1.985		0.19	221.010	C.P.
I			1.545 1.515	0.03		221.040	
Σ	4.78		3.74	4.665	3.625		

checks.

$$\Sigma \text{B.S.} - \Sigma \text{F.S.} = \Sigma \text{Rise} - \Sigma \text{Fall} = \text{Last R.L.} - \text{1st R.L.}$$

- (i) $\Sigma \text{B.S.} - \Sigma \text{F.S.} = 4.78 - 3.74 = 1.04 \text{ m}$
(ii) $\Sigma \text{Rise} - \Sigma \text{Fall} = 4.665 - 3.625 = 1.04 \text{ m}$
(iii) $\text{Last R.L.} - \text{1st R.L.} = 221.04 - 220 = 1.04 \text{ m}$
- } Same.

No-3



It is an anti-clockwise traverse.

Included angle at A = FB of forward line - BB of previous line

$$= \text{FB of } AB - \text{BB of } EA$$

{ Using this formula we shall get interior included angles }

So, Interior included angle at A = $100^\circ 45' - 79^\circ 18' = \boxed{21^\circ 27'}$

Included angle at B = FB of BC - BB of AB

$$= 135^\circ 18' - 281^\circ 48' = -146^\circ 30'$$

$$\cong -146^\circ 30' + 360^\circ$$

$$= \boxed{213^\circ 30'}$$

Angle at C = FB of CD - BB of BC

$$= 45^\circ 12' - 316^\circ 10'$$

$$= -270^\circ 58'$$

$$\cong -270^\circ 58' + 360^\circ = \boxed{89^\circ 2'}$$

5

Angle at D = FB of DE - BB of CD

$$= 300^\circ 36' - 224^\circ 14' = \boxed{76^\circ 22'}$$

Angle at E = FB of EA - BB of DE

$$= 260^\circ 53' - 122^\circ 39' = \boxed{138^\circ 14'}$$

Sum of all interior included angles

$$\angle A + \angle B + \angle C + \angle D + \angle E = 538^{\circ} 35'$$

But, Sum of all angles (interior included angles) should be 540

2.5 Error is $540^{\circ} - 538^{\circ} 35' = 1^{\circ} 25'$ (-ve error)

So the correction will be positive. In each angle $\frac{125'}{5}$

= 17' is to be added.

Corrected angles are.

2.5 $\angle A = 21^{\circ} 27' + 17' = 21^{\circ} 44'$

$\angle B = 213^{\circ} 30' + 17' = 213^{\circ} 47'$

$\angle C = 89^{\circ} 2' + 17' = 89^{\circ} 19'$

$\angle D = 76^{\circ} 22' + 17' = 76^{\circ} 39'$

$\angle E = 138^{\circ} 14' + 17' = 138^{\circ} 31'$

To choose which line is having minimum error

4

Line	FB	BB	FB ~ BB	Error
AB	100° 45'	281° 48'	181° 3'	+1° 3'
BC	135° 18'	316° 10'	180° 52'	+ 52'
CD	45° 12'	224° 14'	179° 2'	- 58'
DE	300° 36'	122° 39'	177° 57'	-2° 3'
EA	260° 53'	79° 18'	181° 35'	+1° 35'

← minimum

Let's choose line BC, as it is having minimum error. The error is distributed in both FB and BB.

Line BC.

2

$$\begin{aligned} \text{Correct FB} &= 135^{\circ} 18' + \frac{52'}{2} = 135^{\circ} 44' \\ \text{Correct BB} &= 316^{\circ} 10' - \frac{52'}{2} = 315^{\circ} 44' \end{aligned} \quad \left. \begin{array}{l} \text{difference} \\ \text{is } 180^{\circ} \end{array} \right\}$$

Line CD

(5)

$$\begin{aligned}
 \text{BB of line CD} &= \text{FB of line BC} + \angle C \\
 &= 135^\circ 44' + 89^\circ 19' \\
 &= 225^\circ 3'
 \end{aligned}$$

(2)

$$\text{FB of line CD} = 225^\circ 3' - 180^\circ = 45^\circ 3'$$

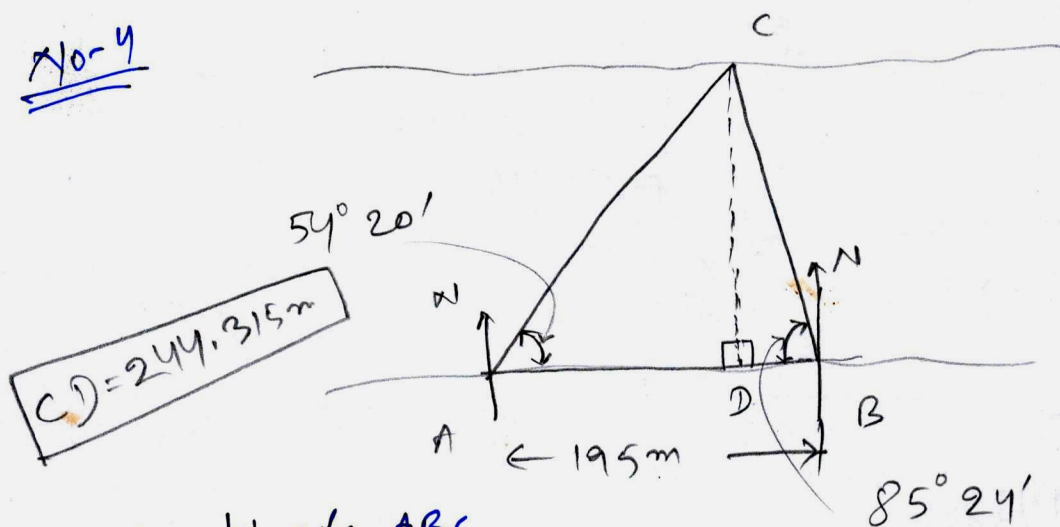
Line DE

$$\begin{aligned}
 \text{BB of line DE} &= \text{FB of line CD} + \angle D \\
 &= 45^\circ 3' + 76^\circ 39' \\
 &= 121^\circ 42'
 \end{aligned}$$

(2)

$$\text{FB of line DE} = 121^\circ 42' + 180^\circ = 301^\circ 42'$$

Q-4



$$\begin{aligned}
 \angle A &= 90^\circ - 35^\circ 40' \\
 &= 54^\circ 20'
 \end{aligned}$$

$$\begin{aligned}
 \angle B &= 355^\circ 24' - 270^\circ \\
 &= 85^\circ 24'
 \end{aligned}$$

In triangle ABC

$$\angle C = 180^\circ - (54^\circ 20' + 85^\circ 24') = 40^\circ 16'$$

Sine rule

(10)

$$\frac{\sin A}{BC} = \frac{\sin B}{AC} = \frac{\sin C}{AB}$$

$$\text{So } \frac{\sin B}{AC} = \frac{\sin C}{AB} \Rightarrow$$

$$\begin{aligned}
 AC &= AB \times \frac{\sin B}{\sin C} \\
 &= 195 \times \frac{\sin (85^\circ 24')}{\sin (40^\circ 16')} \\
 &= 300.724 \text{ m}
 \end{aligned}$$

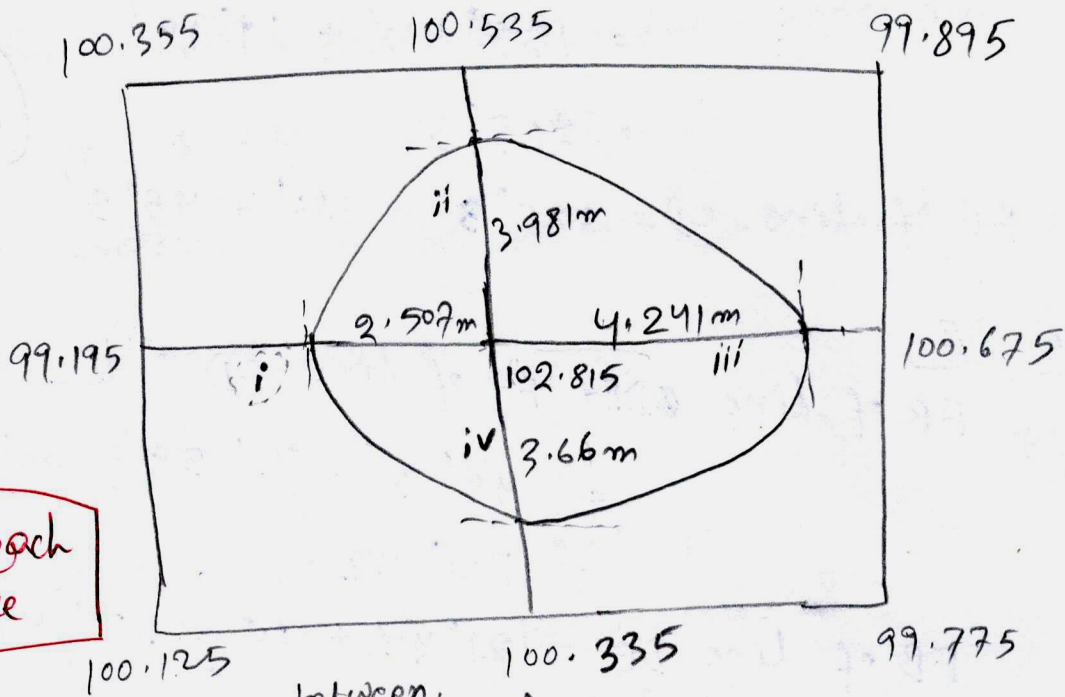
Similarly from triangle ACD

(5)

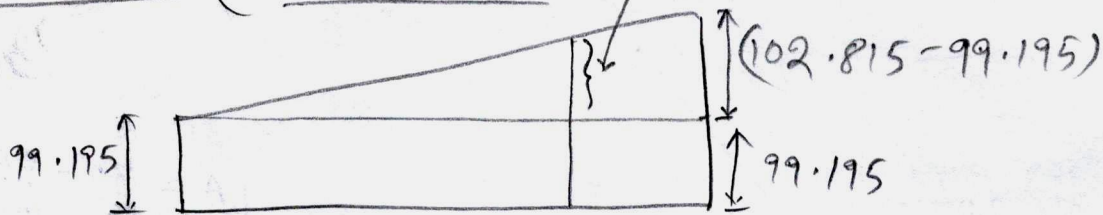
$$\frac{\sin A}{CD} = \frac{\sin D}{AC} \Rightarrow CD = AC \times \frac{\sin A}{\sin D} = 300.724 \times 0.812$$

No-5

10
2.5 for each distance



(i) Calculation. $(99.195 \sim 102.815)$ between. $(101 - 99.195)$



$$\leftarrow (5-x) \longrightarrow \leftarrow x \longrightarrow$$

$$\leftarrow 5 \longrightarrow$$

$$\frac{102.815 - 99.195}{(5)} = \frac{(101 - 99.195)}{(5-x)}$$

$$\Rightarrow \frac{3.62}{5} = \frac{1.805}{5-x}$$

$$\Rightarrow 5-x = 2.493$$

$$\Rightarrow x = 2.507 \text{ m}$$

(ii) $100.535 \sim 102.815$

$$\frac{102.815 - 100.535}{5} = \frac{101 - 100.535}{5-x}$$

$$\Rightarrow \frac{2.28}{5} = \frac{0.465}{5-x} \Rightarrow 5-x = 1.019$$

$$x = 3.981 \text{ m}$$

(7)

$$1) \quad 100.675 \sim 102.815$$

$$\frac{102.815 - 100.675}{5} = \frac{101 - 100.675}{5-x}$$

$$\Rightarrow \frac{2.14}{5} = \frac{0.325}{5-x}$$

$$\Rightarrow 5-x = 0.759$$

$$\Rightarrow \boxed{x = 4.241 \text{ m}}$$

$$(iv) \quad 100.355 \sim 102.815$$

$$\frac{102.815 - \overset{100.385}{\cancel{100.355}}}{5} = \frac{101 - \overset{100.335}{\cancel{100.355}}}{5-x}$$

$$\Rightarrow \frac{2.48}{5} = \frac{0.665}{5-x}$$

$$\Rightarrow 5-x = 1.340$$

$$\Rightarrow \boxed{x = 3.66 \text{ m}}$$

Those who have taken full length ^{sizes} as 5m x 5m is also given full credit.